

Algebra

Quotient Rings & Ideals

$$R \longrightarrow R/I$$

$$\mathbb{Z} \longrightarrow \mathbb{Z}/(12)$$

$$\mathbb{C}[x] \longrightarrow \mathbb{C}[x]/\langle g(x) \rangle$$

$$\mathbb{F}_q[x] \longrightarrow \mathbb{F}_q[x]/\langle x^n - 1 \rangle$$

Quotienting



Closed Subscheme

Ideals of A



Cyclic Codes

Geometry

Affine Schemes & Varieties

$$\text{Spec}(R/I) \cong V(I) \subseteq \text{Spec}(R)$$

Cuts out points (2), (3)

Cuts out roots of g

Cuts out $V(x^n - 1)$

Coding Theory

Vector Spaces & Subspaces

$$A = \mathbb{F}_q[x]/\langle x^n - 1 \rangle \cong \mathbb{F}_q^n$$

Multiplication by x



Cyclic Shift

Roots of generator $g(x)$
define code properties

Structure of $x^n - 1 = g(x)h(x)$

